AI/ML intern DAY 5

TASK 3 — ADVANCED STUDY OF VIRTUAL TRY-ON DATASETS

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1. Abstract

This task focused on researching datasets used in Artificial Intelligence virtual try-on systems and computer vision-based fashion applications. The study explored how AI training datasets improve pose estimation, clothing segmentation, garment alignment, and virtual fitting accuracy.

The research also analyzed how datasets are structured, annotated, and processed for machine learning workflows in digital fashion technology.

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2. Research Objective

The primary objective of this task was to study AI datasets used in virtual try-on systems and understand how machine learning models are trained using fashion-related data.

The research focused on:

- virtual try-on datasets
- pose estimation datasets
- clothing segmentation datasets
- fashion image annotation
- AI training workflows
- body landmark datasets

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3. Introduction to AI Fashion Datasets

AI fashion datasets are structured collections of images and annotations used to train machine learning models for fashion-related computer vision applications.

These datasets help AI systems learn:

- body pose estimation
- clothing recognition
- human segmentation
- garment alignment
- outfit recommendation
- virtual fitting simulation

The datasets generally contain:

- human body images
- clothing images
- segmentation masks
- body landmarks
- pose annotations
- fashion labels

4. Structure of Virtual Try-On Datasets

Most virtual try-on datasets contain multiple data components.

Dataset Component	Purpose
Human Images	Body pose analysis
Clothing Images	Garment simulation
Pose Annotations	Landmark detection
Segmentation Masks	Body region separation
Fashion Labels	Clothing categorization

These components help AI systems perform:

- pose tracking
- cloth alignment
- body segmentation
- clothing recommendation

5. Popular AI Fashion Datasets

Several datasets are widely used in virtual try-on and AI fashion applications.

5.1 DeepFashion Dataset

Overview

DeepFashion is a large fashion dataset used in computer vision and machine learning research. It contains fashion images with clothing labels, pose annotations, and landmark information.

Features

- over 800,000 fashion images
- clothing labels
- pose annotations
- landmark information

Improves

- clothing recognition
- fashion recommendation
- body landmark detection
- virtual fitting systems

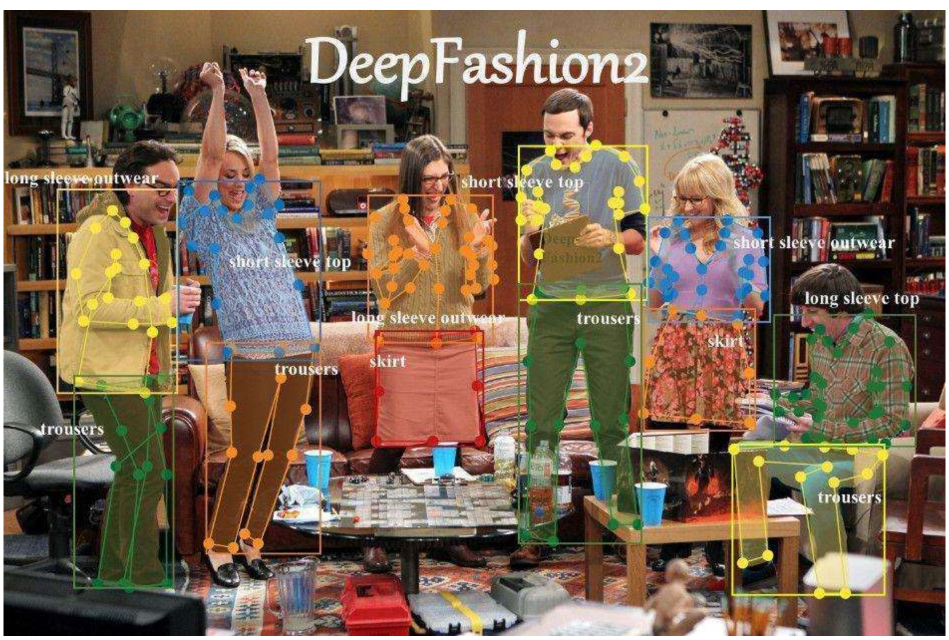

Figure 2 — DeepFashion Dataset

5.2 VITON Dataset

Overview

VITON is a virtual try-on dataset designed for AI clothing fitting and garment transfer systems.

Features

• paired clothing images

• body pose images

• garment alignment samples

• fitting annotations

Improves

• virtual fitting accuracy

• clothing alignment

• garment transfer

• digital clothing simulation

Figure 3 — VITON Virtual Try-On Dataset

5.3 Fashion-MNIST Dataset

Overview

Fashion-MNIST is a lightweight fashion dataset used for image classification and beginner machine learning tasks.

Features

- grayscale clothing images
- labelled fashion categories
- lightweight dataset

Improves

- clothing classification
- fashion image recognition
- AI model training

Figure 4 — Fashion-MNIST Dataset

5.4 ATR Dataset

Overview

ATR is a human parsing dataset used for body segmentation and clothing region analysis.

Features

• body segmentation masks

• clothing labels

• body parsing annotations

Improves

• body segmentation

• clothing region detection

• virtual fitting quality

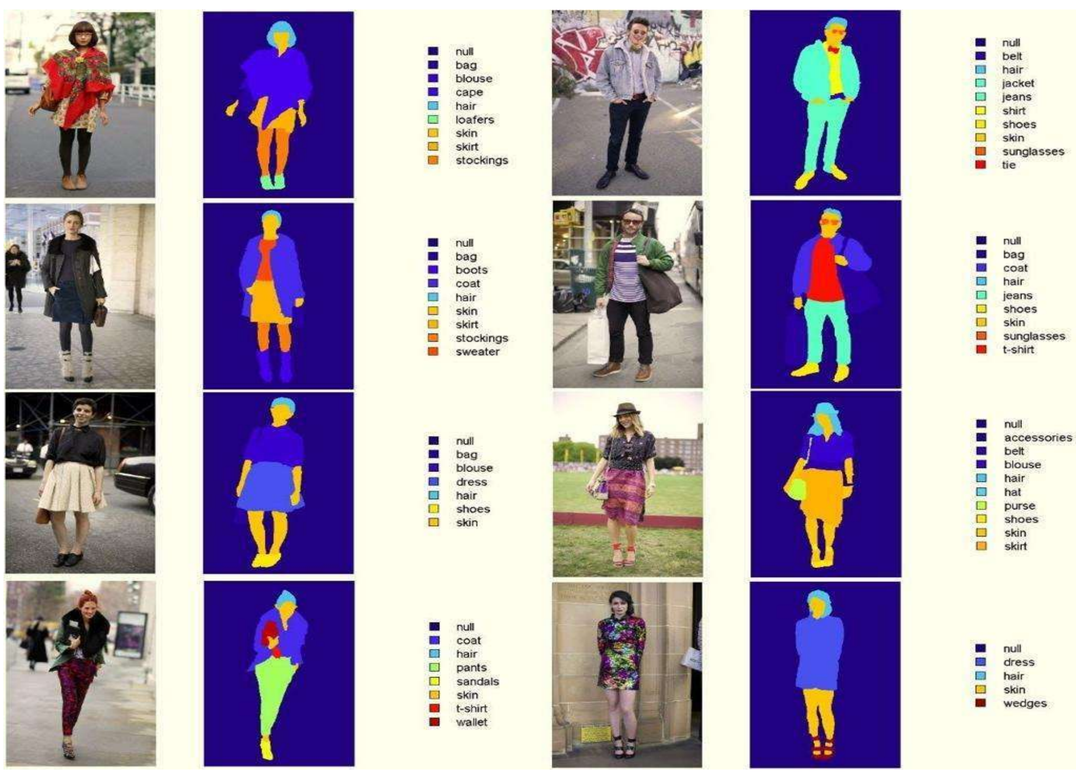

Figure 5 — ATR Human Parsing Dataset

5.5 CIHP Dataset

Overview

CIHP (Crowd Instance-level Human Parsing) is a large-scale human segmentation dataset.

Features

- multiple human parsing
- segmentation masks
- body region annotations
- clothing parsing labels

Improves

- human segmentation
- crowd body analysis
- AI fashion applications
- body tracking systems

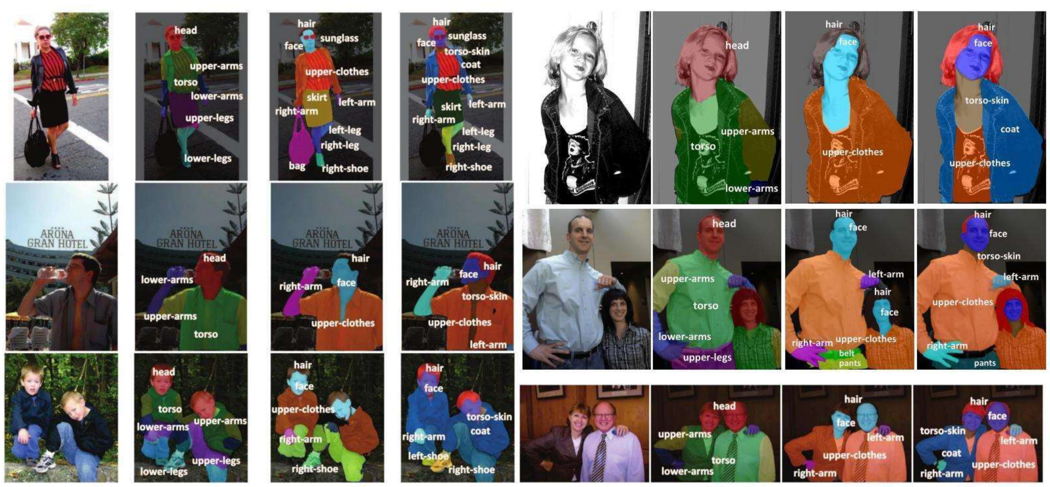

Figure 6 — CIHP Human Parsing Dataset

6. AI Dataset Training Workflow

The AI training pipeline using fashion datasets generally follows several stages.

Step 1 — Data Collection

Fashion images and pose data are collected.

Step 2 — Data Annotation

Human body landmarks and clothing regions are labelled.

Step 3 — Data Preprocessing

Images are resized and normalized for AI training.

Step 4 — Model Training

Machine learning models learn clothing and body patterns.

Step 5 — Testing and Validation

The AI system is tested for prediction accuracy.

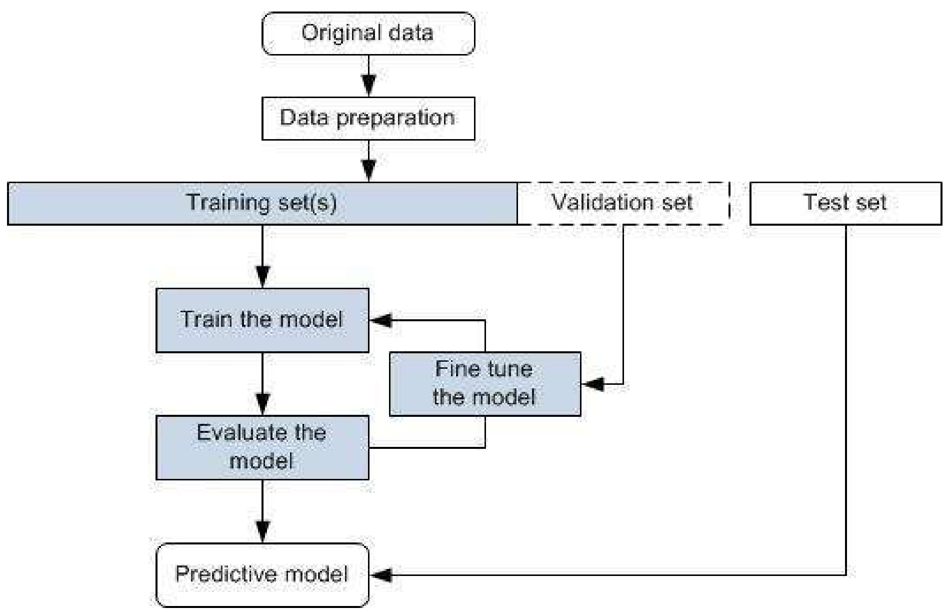

Figure 7 — AI Dataset Training Workflow

6.1 Basic Dataset Verification using Google Colab

A simple dataset loading process was studied using Google Colab to understand how AI systems access and process fashion datasets.

Sample Code

```
from tensorflow.keras.datasets import fashion_mnist

(trainX, trainY), (testX, testY) = fashion_mnist.load_data()

print("Training Data Shape:", trainX.shape)

print("Testing Data Shape:", testX.shape)
```

Observations

- Dataset loading was successful
- Images were properly structured
- Dataset was suitable for machine learning workflows
- Fashion image categories were easy to analyze

7. Advantages of AI Fashion Datasets

The advantages include:

- improved AI prediction accuracy
- realistic garment simulation
- better body segmentation
- efficient AI training
- improved clothing alignment
- enhanced virtual fitting quality

8. Challenges in Dataset Preparation

Challenge	Impact
Large dataset size	High storage requirement
Annotation complexity	Time-consuming labelling
Lighting variations	Affects training accuracy
Pose diversity	Requires multiple body positions
Dataset imbalance	Reduces model performance

9. Knowledge and Skills Acquired

During this research task, the following concepts were understood:

- AI dataset workflows
- fashion image annotation
- pose estimation datasets
- clothing segmentation systems
- machine learning training pipelines
- virtual try-on systems
- AI data preprocessing

10. Conclusion

Virtual try-on datasets are essential for training Artificial Intelligence systems used in modern fashion technology applications.

The research improved understanding of:

- AI fashion datasets
- dataset annotation systems
- clothing segmentation
- pose estimation workflows
- machine learning training
- digital fashion technologies

These datasets play a major role in improving AI virtual fitting systems, intelligent shopping platforms, and computer vision-based fashion applications.